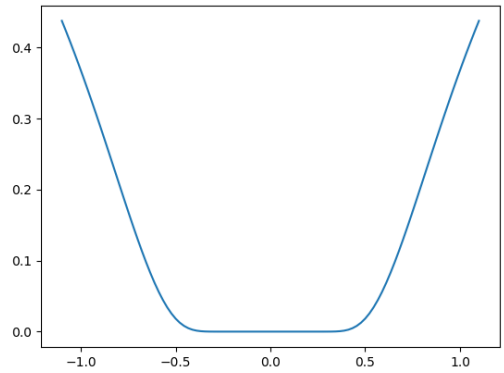
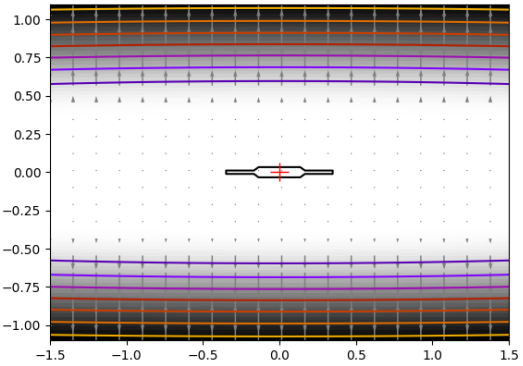
Exercice: A locally flat minimum

Consider the function *exp(-1/(.1*x**2 + y**2)*. This function admits a minimum in (0, 0). Starting from an initialization at (1, 1), try to get within 1e-8 of this minimum point.





13.6 Special case: non-linear least-squares

13.6.1 Minimizing the norm of a vector function

Least square problems, minimizing the norm of a vector function, have a specific structure that can be used in the Levenberg–Marquardt algorithm implemented in `scipy.optimize.leastsq()`.

Lets try to minimize the norm of the following vectorial function:

```
>>> def f(x):
...     return np.arctan(x) - np.arctan(np.linspace(0, 1, len(x)))

>>> x0 = np.zeros(10)
>>> sp.optimize.leastsq(f, x0)
(array([0.        , 0.11111111, 0.22222222, 0.33333333, 0.44444444,
       0.55555556, 0.66666667, 0.77777778, 0.88888889, 1.        ]), 2)
```

This took 67 function evaluations (check it with 'full_output=1'). What if we compute the norm ourselves and use a good generic optimizer (BFGS):